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Groundwater activity on Mars and implications for a deep biosphere

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By the time eukaryotic life or photosynthesis evolved on Earth, the martian surface had become extremely inhospitable, but the subsurface of Mars could potentially have contained a vast microbial biosphere. Crustal fluids may have welled up from the subsurface to alter and cement surface sediments, potentially preserving clues to subsurface habitability. Here we present a conceptual model of subsurface habitability of Mars and evaluate evidence for groundwater upwelling in deep basins. Many ancient, deep basins lack evidence for groundwater activity. However, McLaughlin Crater, one of the deepest craters on Mars, contains evidence for Mg–Fe-bearing clays and carbonates that probably formed in an alkaline, groundwater-fed lacustrine setting. This environment strongly contrasts with the acidic, water-limited environments implied by the presence of sulphate deposits that have previously been suggested to form owing to groundwater upwelling. Deposits formed as a result of groundwater upwelling on Mars, such as those in McLaughlin Crater, could preserve critical evidence of a deep biosphere on Mars. We suggest that groundwater upwelling on Mars may have occurred sporadically on local scales, rather than at regional or global scales.

One of the most important discoveries in the exploration of Mars has been the detection of putative hydrothermal phases, including serpentine[1] and phyllosilicates[2], within materials exhumed from the subsurface by large impact craters[2–5] (Fig. 1). Deep (kilometre-scale) subsurface alteration probably peaked in the Noachian (>4.1 Gyr ago) and into the Early Hesperian (~3.7–4.1 Gyr ago) periods[2], when heat flow was significantly higher[6]. This time period roughly coincides with the earliest record of life on Earth, which consists of prokaryote thermophiles[7] (Fig. 2).

Today, prokaryotic life in the deep subsurface comprises up to 50% of the total biomass on Earth[8]. A significant amount of diversity exists throughout the huge volume of subsurface habitable environments that may reach >5 km depth[9]. As chemoautotrophs and thermophiles are some of the oldest phyla, it stands to reason that life may have originated in the subsurface by taking advantage of existing chemical gradients associated with serpentinization reactions[10], or that thermophiles uniquely survived the Late Heavy Bombardment by taking refuge in the subsurface[11]. The subsurface could have been the most viable habitat for ancient, simple life forms on Mars as well.

Exploration of the habitability of the martian subsurface would provide critical information about geochemical processes in the early history of the Solar System and an essential piece of Earth's geologic puzzle. The investigation of life's origins on Earth will always be limited by the poor state of preservation of the earliest geologic record (>3.5 Gyr old). Therefore, the search for early chemical steps that led to life's origins may ultimately require exploration beyond Earth, specifically characterization of ancient crustal environments on Mars.

Subsurface processes on Mars could be studied indirectly, either by the analysis of deep crustal rocks that have been exhumed by impact or through investigation of materials formed from subsurface fluids[12], where they have reached the surface. Here, we produce a synthesis model of the subsurface geology of Mars, with predictions for the nature and fate of fluids in the crust and testable hypotheses for the habitability of various zones at depth. We also present evidence that crustal fluids have emerged at the surface, resulting in an alkaline lacustrine system within McLaughlin Crater.

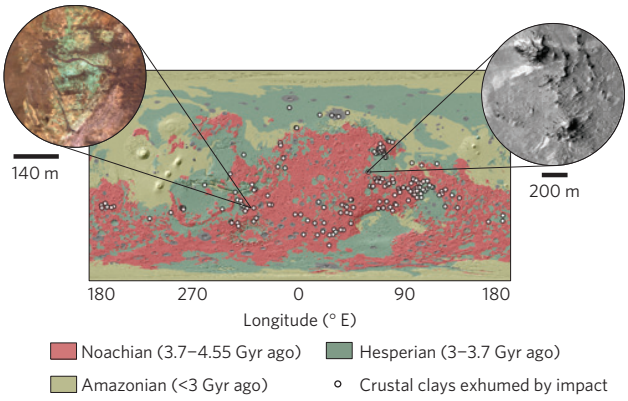


Figure 1 | Distribution of exhumed deep crustal rocks on Mars. Detections of deep crustal clays reported previously[2] are overlaid on global surface geology. Exhumed clays in Noachian terrains represent subsurface hydrothermal processes early in Mars's history. Insets show textures of two examples of exhumed crust: hydrated minerals along with mafic mineralogy exhumed from a ~2.5-km-deep unnamed crater at 306.4° E, 20.5° S (left) and Fe–Mg clays and Fe/Ca carbonates exhumed from ~6 km deep in Leighton Crater (right).

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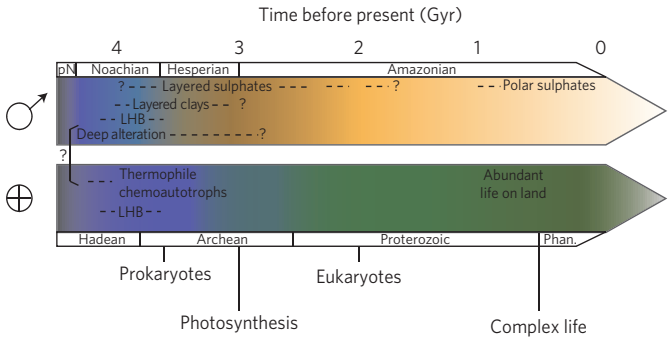


Figure 2 | Major evolutionary events on Earth compared with geologic evolution of Mars. Both planets experienced major hydrothermal episodes before and during the Late Heavy Bombardment. At the time when the first known life on Earth existed, Mars was habitable, but the martian surface became hyperarid, extremely oxidized, and inhospitable before the evolution of photosynthesis or eukaryotic life on Earth. The martian subsurface is the place to search for evidence of habitability. On Mars, pN is the Pre-Noachian.

Geology of the martian subsurface

Water volcanically outgassed during the Pre-Noachian and Noachian periods on Mars would have ultimately become locked into the cryosphere, sequestered into the subsurface, and/or lost to space[13]. The modern observable cryosphere contains an equivalent global layer of water (EGL) ~35 m thick[14], which has periodically been mobilized throughout the surface through obliquity-driven climate change[15–17], and probably has recharged the subsurface through basal melting[18]. However, the largest fraction of martian H_2O probably occurs in the subsurface as briny groundwater[19], a deep cryosphere[18], and water in hydrated minerals[2].

Water that infiltrated into the subsurface would have met one of several fates as pore water, ground ice, or structural and absorbed water in hydrated minerals. Owing to its reduced gravity, Mars contains more subsurface porosity to a greater depth than Earth[18]. For example, a saturated zone at a depth >6 km below the surface could account for an EGL of water ~100 m thick, and a saturated zone below 3 km depth would amount to an EGL of 300 m (Fig. 3). Ancient fluids that arrived in the saturated zone during periods of relatively high recharge rates on ancient Mars are likely to have become dense owing to a high concentration of dissolved solids[19], acquired during transport through reactive, mafic rocks. Even if average surface temperatures were 10 s of degrees below freezing, hydrothermal conditions could have existed at depths of several kilometres during the Noachian, given reasonable estimates for ancient thermal gradients[6] (Fig. 3).

The alteration mineralogy at depth can be determined from the composition of impact craters, which provide natural probes into the subsurface (in particular, the uplift zones in central peaks). Alteration minerals are more common in materials exhumed by craters >20 km diameter[3]. Given that the scaling relation of exhumation depth to final diameter is ~1:10 for large craters[20], it may be inferred that subsurface alteration is relatively higher at depths >2 km. In fact, a large fraction of the most intensely altered central peaks of impact craters occur in craters of 50–100 km diameters[2], which implies the most intense alteration at depths >5 km. In our conceptual view of martian subsurface geology, we refer to this deep (>5 km), hydrothermal, fluid-rich region as zone 4 (Fig. 3). Fluids in this zone may have become so dense and deep that they rarely, if ever, re-emerged at the surface.

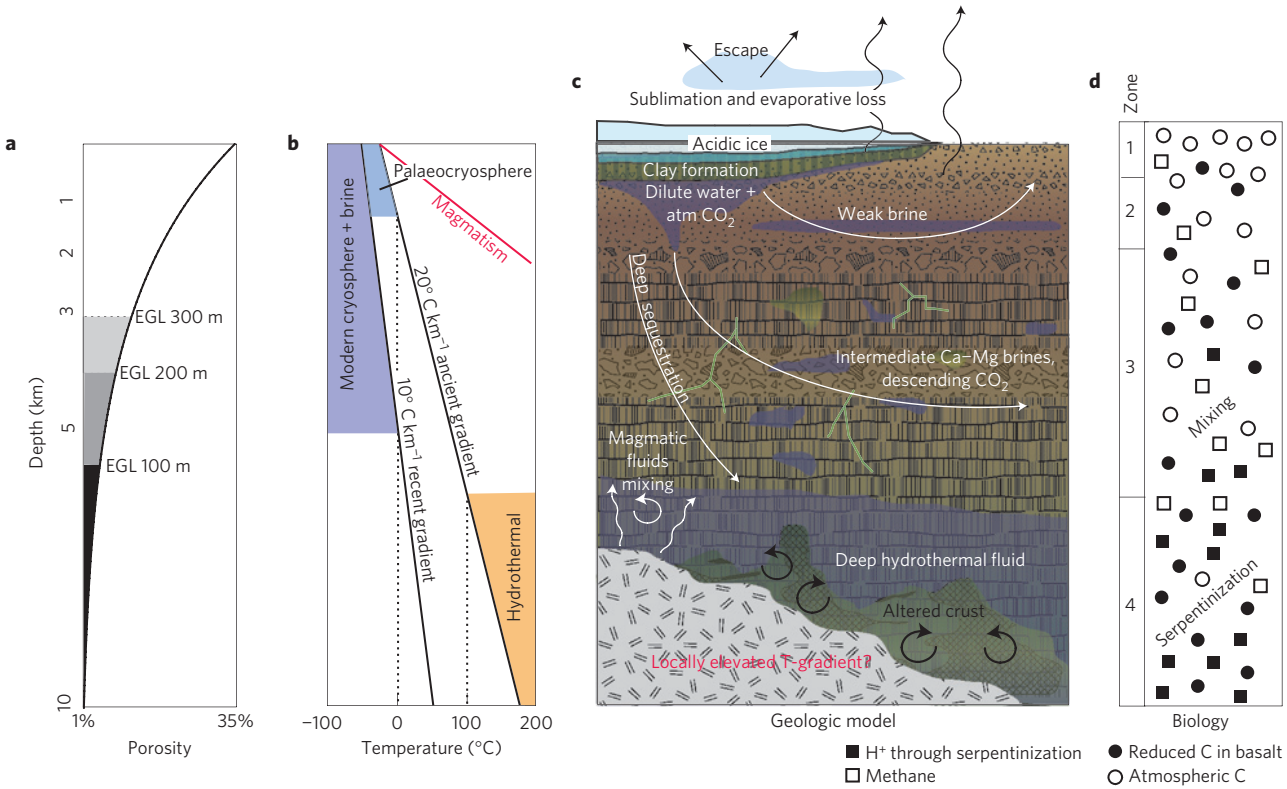


Figure 3 | Synthesis model of subsurface geology and habitability on Mars at indicated depths. a, Decay of porosity in the martian crust[18], where EGL is the amount of water than can be stored by saturation. **b**, Two possible thermal gradients (10 and 20 K km^{-1}) show that hydrothermal temperatures were probably reached at depths several kilometres below the surface. **c**, A schematic diagram shows the geologic contexts of groundwater on Mars, where upper crustal rocks (sediments, impact breccia and volcanics, brown; igneous intrusions, grey) have been altered to clays (green) by fluids (purple). Blue denotes surface ice layers. **d**, The distribution of energy sources and nutrients for microorganisms in the subsurface for the zones discussed in the main text.

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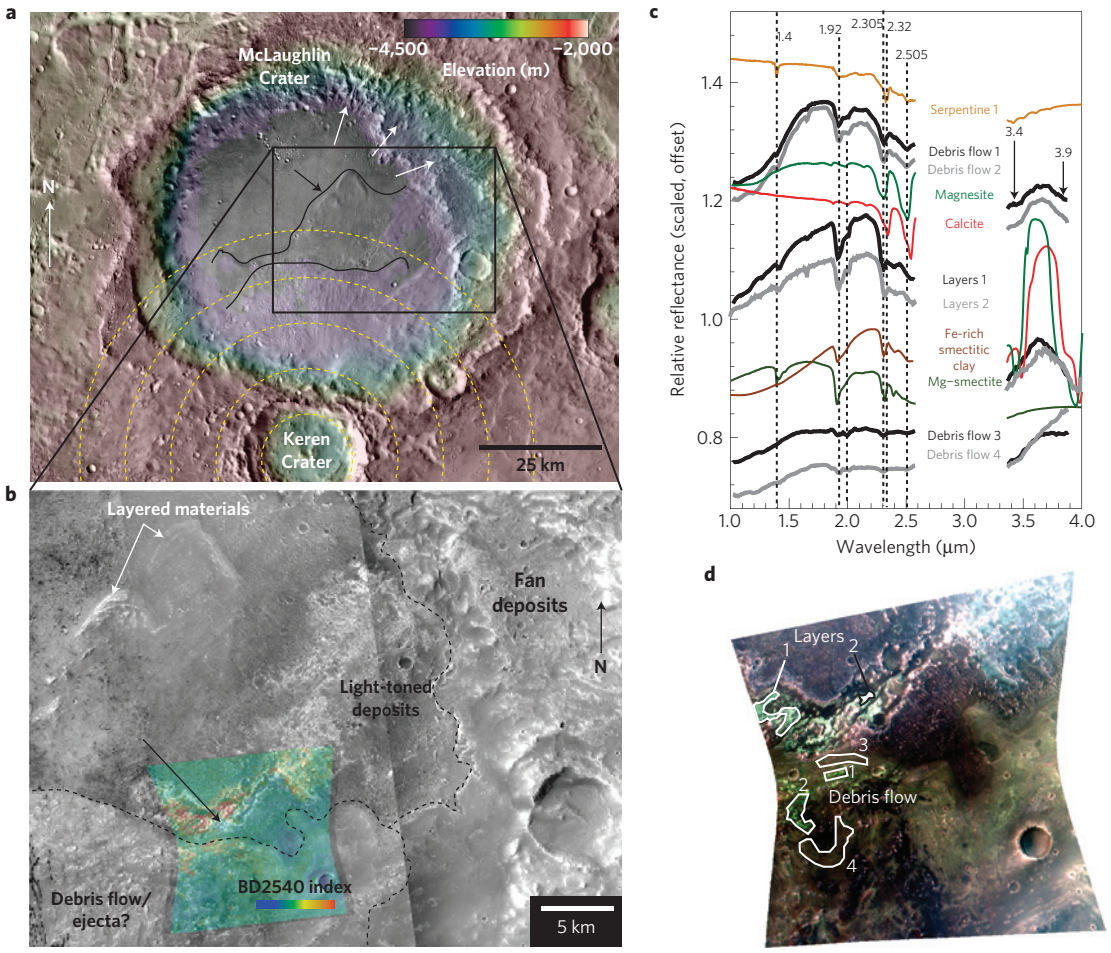


Figure 4 | Mineralogy and geomorphology of McLaughlin Crater. MOLA data are shown in **a** with channels indicated by white arrows and possible debris flows marked by black lines and arrows. Yellow lines mark crater radii from Keren Crater. **b**, CTX data show the occurrence of light-toned deposits and putative debris flows on the crater floor, as well as the location of CRISM image AA5A (**d**). The BD2540 index measures absoprtion (warm colours) related to carbonate and clay minerals. **c**, CRISM spectra extracted from regions of interest in **d** are shown as offset ratio spectra compared to laboratory spectra of relevant minerals.

Materials exhumed from 2 to 5 km depth commonly contain alteration minerals[3] along with unaltered mafic minerals[21] indicating the presence of a large unsaturated, partially altered zone at these depths, which we term zone 3. Water descending to this depth might ultimately be completely consumed by hydrolysis and hydration reactions with rocks. In addition, modest amounts of groundwater could become widely disseminated in the pore space and ultimately seized by capillary forces. This zone might have been active only early in Mars's history when recharge rates were probably much higher. If fluids were able to transit large fractions of the crust and re-emerge in groundwater upwelling zones, the depth that the fluids reached would be proportional to their transit distance and related chemical evolution. Thus, fluids that emerged from zone 3 should have had neutral to alkaline pH and been enriched in silica, Ca^{2+}, Mg^{2+} and Fe^{2+}, owing to long transport paths in the subsurface (Fig. 3).

The shallow subsurface is relatively unaltered, and consists largely of basaltic materials to depths of ~2 km (ref. 22). Zone 2 is the region of the crust where hydrothermal temperatures may have never been reached, except locally owing to magmatic intrusion. Relatively dilute brines might have formed in this region, becoming enriched in the most mobile cations such as K^+ and Na^+. If precipitation early in Mars's history produced fluids that existed in this zone, they might have re-emerged after short traverse distances and combined with volcanogenic Cl^- to precipitate Na and K chlorides in depressions and breaks in slope where the shallow fluids could reach the surface. Such a scenario is consistent with the topographic settings of chloride deposits observed throughout the ancient crust[23].

Last, the surface layer, which we term zone 1, contains sulphates[24] and layered clays[25], as well as mobile surface sediments, snow and ice deposits, and dust[26]. This zone may occupy the uppermost hundreds to thousands of metres of the surface of many regions of Mars[27] (Fig. 3). Fluids in this zone would have been largely affected by interaction with atmospheric SO_2, Cl^- and oxidizing agents[28].

Possibility of a deep biosphere

The distinction of these geological zones within the crust allows us to predict how the potential for a subsurface martian biosphere[29] varies with depth. In the deepest zone (zone 4), the viability of a microbial community is perhaps greater than at similar depths on Earth because a lower gravity implies less compaction of the very limited pore space, and lower heat flow reduces the temperature constraints (Fig. 3)[30]. At this depth, a diversity of metabolic mechanisms may be expected, using hydrogen, carbon dioxide and possibly abiotic hydrocarbons, but we assume that as on Earth the dominant fuel for microbes in the deep subsurface would be hydrogen[31], possibly provided by serpentinization[32], radiolysis[33] or fault friction[34]. Carbon for biomass could have been derived from the

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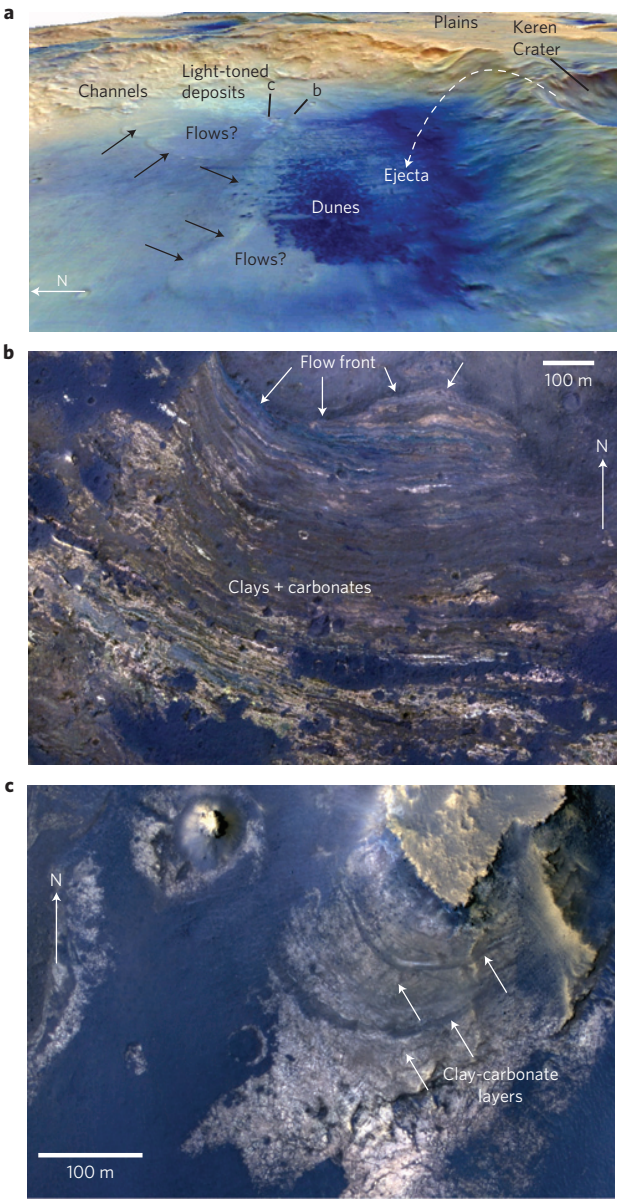


Figure 5 | Geology of McLaughlin Crater. a, Colour image data from the High Resolution Stereo Camera (HRSC) draped onto elevation data from the same instrument show the locations of features discussed in the text, including possible flow fronts indicated by black arrows. The locations of **b**,**c** are indicated. **b**,**c**, Contain image data from the High Resolution Imager for Mars (HiRISE) showing altered sediments within lobate flows (**b**) and layered rocks in the floor of the crater (**c**).

magmatic carbon in basalts (Fig. 3) whose occurrence is observed in martian meteorites[35]. In zone 3, CO_2-bearing fluids infiltrated from the surface could have reacted with the hydrogen from serpentinization to form methane, which could have fuelled methanotrophs and been mediated by methanogens. The alteration of basalt also releases cations, for example Fe^{2+} and Mg^{2+}, which can fuel iron respiration[36] and facilitate the formation of complex organic molecules[37], respectively. Groundwater recharge and upwelling events provide mechanisms for replenishment of nutrients, as well as possible redox-based habitats at the interface of seepage pathways and host rocks, as is observed to be an important control on Earth[38]. The combination of recharging fluids and serpentinization can give rise to unusual alkaline (Ca/Mg-rich) springs with distinctive microbial communities, proposed as a potential model for Mars[39].

The deep crust may have always been the most habitable environment on Mars. Zone 4 would have been energy-rich, but potentially more nutrient-limited whereas zone 3 could have included diverse chemistry in a context with ample energy and nutrients. Simple life could have evolved into habitats in zones 1 and 2. Zone 2, at shallower depth and with less alteration, offers porous habitats protected from harsh surface conditions. These specifically might include vesicular basalts, which are a widely colonized habitat on Earth, and fracture systems associated with impact hydrothermal systems, also colonized in the terrestrial geological record[40]. Zone 1 offers other targets that are inhabited on Earth, including ice with brine inclusions[41], sulphates and layered clays.

Exploring for a deep biosphere on Mars

Present exploration of Mars is focused on evaluating the habitability of surface environments. However, the martian surface has been extremely cold, oxidizing and arid through most of its history, long before the evolution of eukaryotic life forms on Earth and for billions of years before the earliest evidence of surface life on land (Fig. 2). We argue that the highest priority astrobiological targets on Mars should be portions of deep crust exhumed by impact and erosion, which could preserve evidence of organic chemicals from an era that is not preserved in Earth's geologic record. Alternatively, high-priority sites would also be areas where fluids have emerged from the martian subsurface, possibly carrying clues to subsurface geochemistry and prebiotic or biotic processes. In fact, sites of groundwater upwelling might provide chemical gradients advantageous to life.

Groundwater upwelling on Mars

Despite the fact that the martian crust is probably hydrologically heterogeneous owing to impact fragmentation and other processes, theoretical groundwater models provide guidance as to how and where groundwater upwelling might have occurred given various model inputs[12]. These models predict that upwelling would have occurred first in deep basins[12]. It is proposed that a rising groundwater table would serve to alter and cement sediments available in those upwelling zones, resulting in deposits of sedimentary rocks that are relics of ancient groundwater activity[12]. The northern hemisphere of Mars contains 95 large (diameter $\geq$40 km), deep (rim-floor depth $\geq$2 km) impact basins[42], approximately 80% of which are Noachian in age (see Supplementary Information), therefore predating predicted upwelling activity[12]. At least one of these basins contains strong evidence for groundwater activity.

Evidence for a groundwater-fed lake in McLaughlin Crater

McLaughlin Crater is a large (diameter = 92 km), deep (2.2 km) crater located at 337.6° E, 21.9° N, adjacent to the global topographic dichotomy boundary. The facts that the crater is deep and situated at a major decline in regional topography suggest that this basin is an excellent candidate in which to search for groundwater activity. Spectra acquired of the floor of McLaughlin Crater by the Compact Reconnaissance Imaging Spectrometer for Mars (CRISM), corrected for instrument and atmospheric effects, contain absorptions at (λ =)1.4, 1.9 and 2.3 μm that are attributable to Fe–Mg-rich clay minerals (Fig. 4). Specifically, the spectral shapes of the floor materials are most similar to those of Fe–Mg–smectite and possibly serpentine (Fig. 4). These same spectra also contain absorptions at 3.4 and 3.9 μm that are diagnostic of carbonates[43]. Absorptions at 2.305 and 2.5–2.52 μm are also consistent with the presence of Mg-carbonates, probably mixed with clay minerals. A spectral index tuned to identify clay minerals with Thermal Emission Spectrometer data[44] also shows the occurrence of alteration minerals within McLaughlin Crater. Analysis of surface emissivity spectra from the Thermal Emission Spectrometer suggests only low abundances (10–15% at scales

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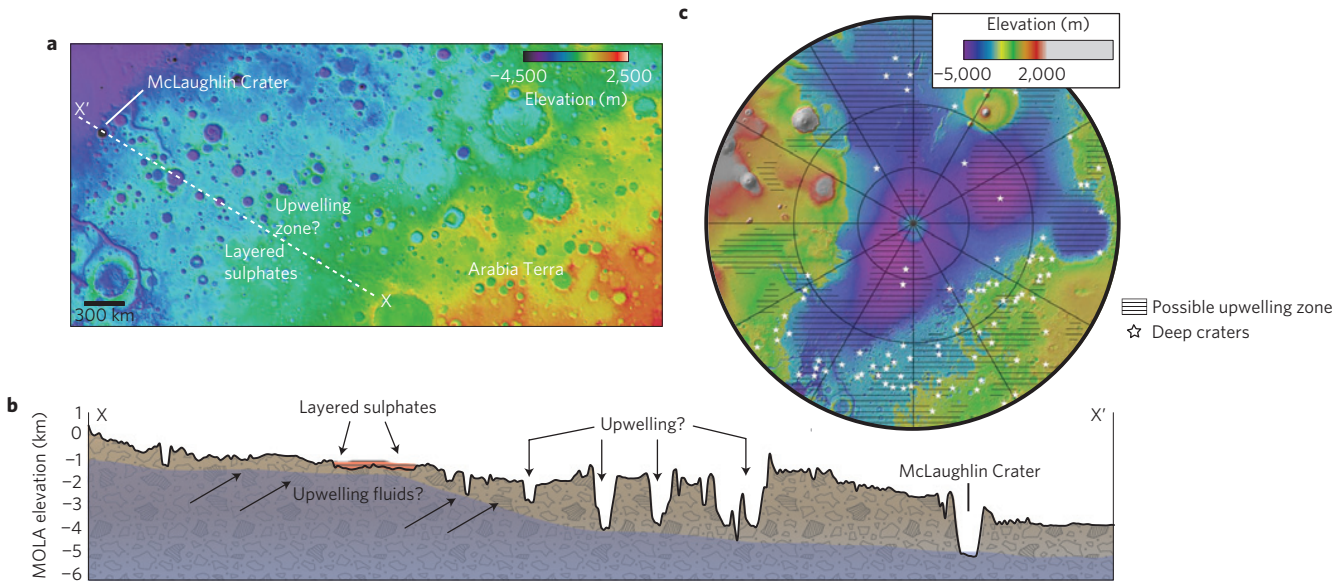


Figure 6 | A survey of evidence for groundwater upwelling in deep craters. a, MOLA data show the Arabia Terra region where layered sulphates proposed to have formed from groundwater upwelling occur. **b**, Cross-section X–X′ illustrates that layered sulphates occur at relatively high elevations, outside basins. Ancient, deep basins in the region should have also experienced groundwater upwelling. **c**, Stars indicate large, deep craters in the northern hemisphere that might be candidate sites for groundwater upwelling activity.

of hundreds of square kilometres) of clays or carbonates in the intracrater sediments (see Supplementary Information).

Channels emanating from the eastern wall of the crater terminate at an elevation ~500 m above the crater floor, which may indicate the former presence of a lake surface (base level) at this elevation (Fig. 4a,b). The channels terminate at a broad platform probably formed from deposition of sediment into standing water from fluvial erosion of the crater wall. The floor of the crater contains layered, flat-lying clay-carbonate-bearing rocks interpreted as lacustrine deposits, which are overlain by deformed, layered, lobate materials thickening southwards towards the southern wall of the crater (Fig. 5).

These lobate materials are important because their presence is probably an indication of rapid burial of crater floor sediments, which is favourable for preservation of organic materials. We interpret the lobate materials as either landslide deposits formed from failure of the southern wall of McLaughlin Crater (Fig. 5) or altered ejecta from Keren Crater on McLaughlin Crater's south rim. The lobate materials are tens of metres thick at distances >5 crater radii from Keren and hundreds of metres thick at a distance of ~3 crater radii; both are an order of magnitude thicker than would be expected on the basis of the modelled thickness of ballistically emplaced ejecta from Keren (see Supplementary Information). The downhill topography could have resulted in greater than expected thicknesses of ejecta towards the north. However, the floor of McLaughlin Crater also contains a number of lobate forms with low relief that probably represent multiple events (Fig. 5). Their lobate, complex morphology suggests a volatile-rich emplacement mechanism (Fig. 5) and therefore, we propose that the materials represent a combination of wet gravity flows and fluidized ejecta emplaced rapidly on the crater floor. On Earth, such geometries are achieved perhaps uniquely in subaqueous landslides[45]. The deposits in McLaughlin Crater could have very high preservation potential for organic materials, in much the same manner as turbidites do on Earth[46].

Taken together, the observations in McLaughlin Crater suggest the basin contained an ancient lake in one of the most likely settings where groundwater would have emerged. The evidence for alteration minerals rich in Fe and Mg, and the indication of alkaline fluids based on carbonate detections, are consistent with the expected character of fluids that would have emerged from the subsurface, perhaps from zones 3 or 4 outlined above.

Rarity of groundwater upwelling events

Although McLaughlin Crater presents a compelling case for groundwater upwelling, on the basis of its depth, regional setting, geomorphology and mineralogy, the style and manifestation of this process is quite different in this crater from the sulphate-rich sedimentation that has been attributed to groundwater upwelling throughout Arabia Terra and elsewhere[12]. Modelled patterns of upwelling provide a reasonable match to the occurrences of sulphate deposits on the plains of Meridiani Planum, if the original depth to a water table was relatively shallow (~600 m-below-surface)[12]. However, the models that predict groundwater upwelling in the plains of Arabia Terra also predict that kilometres of groundwater-driven sedimentation should be found in deep basins throughout the region. Some deep craters do contain interior mounds of sediments that might be attributable to groundwater-mediated sedimentation. Yet, more than 80 deep, ancient basins throughout the northern hemisphere do not exhibit the same evidence. The question arises: how did groundwater activity result in cementation of vast amounts of sediments throughout the plains of Arabia Terra, yet fail to produce evidence for similar deposits in so many deep, ancient basins (Fig. 6)?

There are several possible explanations for the lack of thick, cemented crater fill in deep basins. One possibility is that such deposits once existed, but have since been removed by erosion[47]. However, this scenario is unlikely because it implies higher than expected erosion rates[48] and selective erosion of kilometres of material in some craters and not others. Alternatively, upwelling may have occurred in those basins as predicted by theoretical models, but cemented sedimentary deposits never formed because there was no supply of sediment available in those basins. However, there is no reason to expect that sediment supply would be different in the locations of these basins than elsewhere in Arabia Terra. A third possibility is that upwelling occurred only in Meridiani Planum and several other local settings owing to the heterogeneous nature of the martian crust or to the occurrence of local or regional scale recharge.

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Last, another important possibility is that sulphate-bearing sediments in Meridiani Planum and elsewhere were formed from a water-limited, top-down process that did not require substantial groundwater[49,50]. This would indicate that the lacustrine activity in McLaughlin crater represents the typical manifestation of ancient groundwater processes on Mars, which were much smaller scale and locally heterogeneous than the global processes that have been invoked to explain the substantial sedimentation observed across the equatorial regions of Mars. Lacustrine clay minerals and carbonates in McLaughlin Crater might be the best evidence for groundwater upwelling activity on Mars, and therefore should be considered a high-priority target for future exploration.

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Author contributions

J.R.M. conceived of project, processed most of the data, and wrote most of the manuscript. J.P., P.B.N. and J.C. wrote portions of the paper. A.D.R. contributed analyses of the thermal infrared data and S.P.W. contributed analyses of impact deposits.

Additional information

Supplementary information is available in the online version of the paper. Reprints and permissions information is available online at www.nature.com/reprints. Correspondence and requests for materials should be addressed to J.R.M.

Competing financial interests

The authors declare no competing financial interests.

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